TASK 2 — SQL JOINS

Q1) Task A — INNER JOIN

List employees assigned to projects (only matched records).

Query:

```
SELECT e.employee_id,
       e.employee_fname,
       e.employee_lname,
       p.project_name
FROM employee e
INNER JOIN team t
   ON e.team_id = t.team_id
INNER JOIN project p
   ON t.project_id = p.project_id;
```

Explanation:
This query uses INNER JOINs to retrieve only employees who are assigned to teams that are linked to projects. Employees without matching project records are excluded from the results.

employee(+) 1

SELECT e.employee_id, e.employee_fname, | Enter a SQL expression to filter results (use Ctrl+Space)

	employee_id	employee_fname	employee_lname	project_name
1	1,001	Amit	Sharma	Website Redesign
2	1,002	Riya	Patel	Mobile Application
3	1,003	Rahul	Verma	ERP Deployment
4	1,004	Neha	Iyer	Cloud Migration
5	1,005	Karan	Mehta	Data Warehouse
6	1,006	Priya	Nair	Cybersecurity Audit
7	1,007	Suresh	Reddy	CRM Upgrade
8	1,008	Ananya	Sen	AI Chatbot
9	1,009	Vikram	Singh	E-commerce Portal
10	1,010	Pooja	Gupta	Payment Integration
11	1,011	Arjun	Kapoor	HR Automation
12	1,012	Mehul	Choksi	Inventory System
13	1,013	Tanvi	Kulkarni	Analytics Dashboard
14	1,014	Rakesh	Mishra	Networking Upgrade
15	1,015	Nitin	Aggarwal	IoT Monitoring
16	1,016	Sonal	Desai	Digital Marketing
17	1,017	Mohit	Malhotra	Payroll System
18	1,018	Ayesha	Khan	Document Management
19	1,019	Rohan	Patil	Video Streaming App
20	1,020	Deepak	Yadav	Smart Parking
21	1,021	Kavya	Rao	Banking App
22	1,022	Siddharth	Menon	Telemedicine Platform

Q2) Task B — LEFT JOIN (handle unmatched records)

List all employees and any project they are working on (NULL if none).

Query:

```
SELECT e.employee_id,
       e.employee_fname,
       e.employee_lname,
       p.project_name
FROM employee e
LEFT JOIN team t
   ON e.team_id = t.team_id
LEFT JOIN project p
   ON t.project_id = p.project_id;
```

Explanation:
This query returns all employees regardless of project assignment. The LEFT JOIN ensures that employees without projects still appear in the result set, with NULL values shown for the project name.

	employee_id	employee_fname	employee_lname	project_name
1	1,001	Amit	Sharma	Website Redesign
2	1,002	Riya	Patel	Mobile Application
3	1,003	Rahul	Verma	ERP Deployment
4	1,004	Neha	Iyer	Cloud Migration
5	1,005	Karan	Mehta	Data Warehouse
6	1,006	Priya	Nair	Cybersecurity Audit
7	1,007	Suresh	Reddy	CRM Upgrade
8	1,008	Ananya	Sen	AI Chatbot
9	1,009	Vikram	Singh	E-commerce Portal
10	1,010	Pooja	Gupta	Payment Integration
11	1,011	Arjun	Kapoor	HR Automation
12	1,012	Mehul	Choksi	Inventory System
13	1,013	Tanvi	Kulkarni	Analytics Dashboard
14	1,014	Rakesh	Mishra	Networking Upgrade
15	1,015	Nitin	Aggarwal	IoT Monitoring
16	1,016	Sonal	Desai	Digital Marketing
17	1,017	Mohit	Malhotra	Payroll System
18	1,018	Ayesha	Khan	Document Management
19	1,019	Rohan	Patil	Video Streaming App
20	1,020	Deepak	Yadav	Smart Parking

Q3) Task C — SELF JOIN / COMPLEX JOIN (Bonus task)

Find pairs of employees who belong to the same department.

Query:

```
SELECT e1.employee_fname AS Employee1,
       e2.employee_fname AS Employee2,
```

```
    d.dept_name
FROM employee e1
JOIN employee e2
   ON e1.dept_id = e2.dept_id
  AND e1.employee_id < e2.employee_id
JOIN department d
   ON e1.dept_id = d.dept_id;
```

Explanation:
This query performs a self join on the employee table to identify pairs of employees working in the same department. The condition prevents duplicate and self-matching rows, and the department name is included for clarity.

	employee1	employee2	dept_name
1	Aarav	Zara	Human Resources
2	Varun	Aarav	Human Resources
3	Varun	Zara	Human Resources
4	Amit	Varun	Human Resources
5	Amit	Aarav	Human Resources
6	Amit	Zara	Human Resources
7	Bhavya	Adarsh	Finance
8	Sneha	Bhavya	Finance
9	Sneha	Adarsh	Finance
10	Riya	Sneha	Finance
11	Riya	Bhavya	Finance
12	Riya	Adarsh	Finance
13	Chirag	Binit	Sales
14	Abhishek	Chirag	Sales
15	Abhishek	Binit	Sales

Q4) Task D — FULL JOIN (Departments and Employees)
Shows unmatched records on both sides.

Query:

```
SELECT
   d.dept_name,
   e.employee_fname,
   e.employee_lname
FROM department d
FULL JOIN employee e
   ON d.dept_id = e.dept_id;
```

Explanation:
This query uses a FULL JOIN to retrieve all records from both the department and employee tables. It displays departments that have no employees as well as employees who are not assigned to any department.

When no matching record exists on either side of the join condition, the corresponding columns return NULL values.

	dept_name	employee_fname	employee_lname
1	Human Resources	Zara	Ali
2	Human Resources	Aarav	Shukla
3	Human Resources	Varun	Khanna
4	Human Resources	Amit	Sharma
5	Finance	Adarsh	Nigam
6	Finance	Bhavya	Mittal
7	Finance	Sneha	Joshi
8	Finance	Riya	Patel
9	Sales	Binit	Choudhary
10	Sales	Chirag	Bhardwaj
11	Sales	Abhishek	Tiwari
12	Sales	Rahul	Verma
13	Marketing	Charul	Mathur
14	Marketing	Disha	Arvind
15	Marketing	Pallavi	Kuldeep
16	Marketing	Neha	Iyer
17	Operations	Dev	Ahlawat
18	Operations	Eshan	Malik
19	Operations	Harsh	Bansal